RESP-100 · BASIC EKG INTERPRETATION · WEEKS 12–13

Pacemakers (Paced Rhythms)

Walraven, *Basic Arrhythmias* · Appendix E · Lecture by Michael V Hyland (29 min, posted for Weeks 12 and 13)

In one line: find the vertical **pacer spike**, see **where** it sits, and name the rhythm **A paced**, **V paced** or **AV paced**.

1 · What you need to know | 2 · Pacemaker basics (not tested) | 3 · The pacer spike | 4 · Why V-paced beats are wide | 5 · AV paced | 6 · A paced | 7 · V paced | Cheat sheet | Practice quiz

1 What you actually need to know

Big idea: "This is EKG interpretation, not a pacemaker course." You must be able to say (1) whether a rhythm is **paced**, and (2) whether it is **A paced, V paced or AV paced**. Then fill in the usual 6 blanks.

✔ Tested

- Spotting the vertical **pacer spike**
- Naming the chamber(s) paced from the spike's **location**
- Regularity, rate, P waves, PRI and QRS for paced strips
- Only **3 paced rhythms**: A paced, V paced, AV paced

✘ Not tested (he said so)

- Definitions and reasons for pacing
- Pacemaker parts, types and placement
- Capture, chambers paced, the 3-letter code
- "Smart" sensing and response modes

WHAT THE INSTRUCTOR STRESSED (0:18:26 AND 0:27:43)

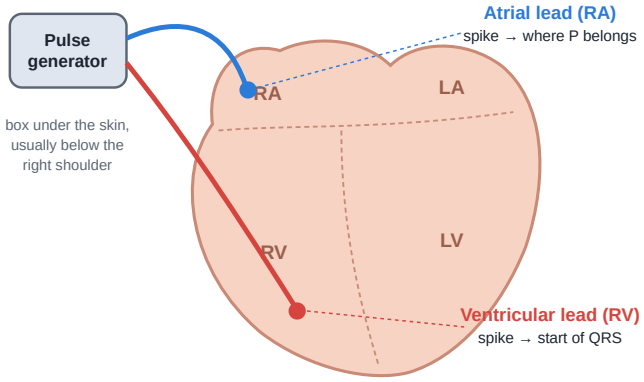
"There **will be an AV paced rhythm on the test**," and definitely on the final. The last homework has 2–3 paced rhythms, plus heart blocks and ventricular rhythms. The next hourly exam covers **ventricular rhythms, heart blocks and paced rhythms**.

FROM CLINICAL

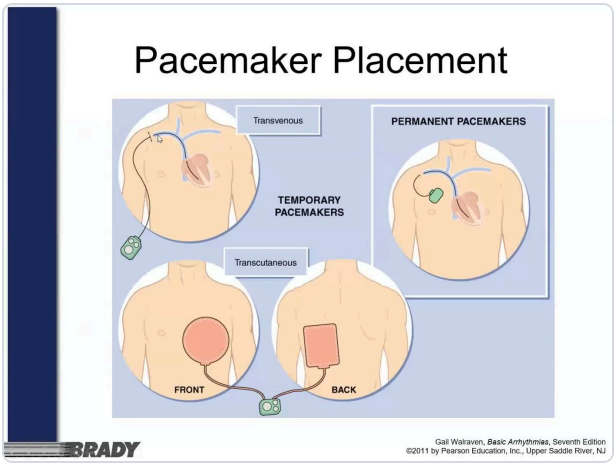
On telemetry monitors, paced rhythms are often colour-coded **purple**. You know a beat was paced because of the **vertical spike** on the screen.

2 Pacemaker basics (background, not tested)

Big idea: A pacemaker is a device that delivers an **artificial electrical stimulus** to make the heart depolarize when the heart's own pacemaker is **too slow**, or when conduction is **blocked** (third-degree heart block).

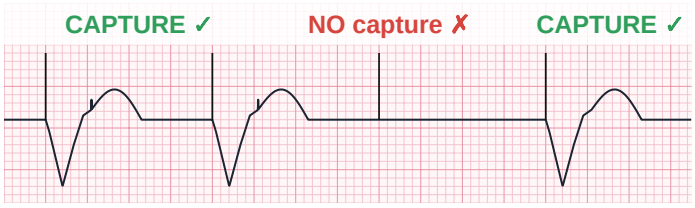


A dual-chamber pacemaker: one lead in the right atrium and one in the right ventricle. The lead that fires decides where the spike appears.

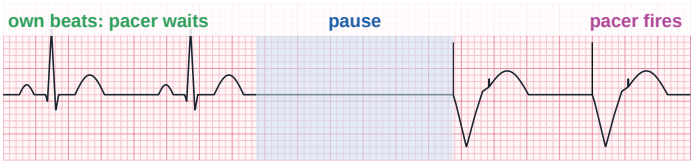


Placement: transvenous and transcutaneous (temporary), and implanted (permanent).

Type	How it works	Use
Transvenous	A wire is threaded through a central IV line (under the collarbone) until it touches the wall of the right ventricle	Temporary, emergency
Transcutaneous	Big pads on the front and back of the chest; the stimulus goes *across the skin* ("cutaneous" = skin)	Temporary, emergency
Permanent (implanted)	A box like a "deck of cards" under the skin near the right shoulder, with wires attached to the heart muscle	Long term



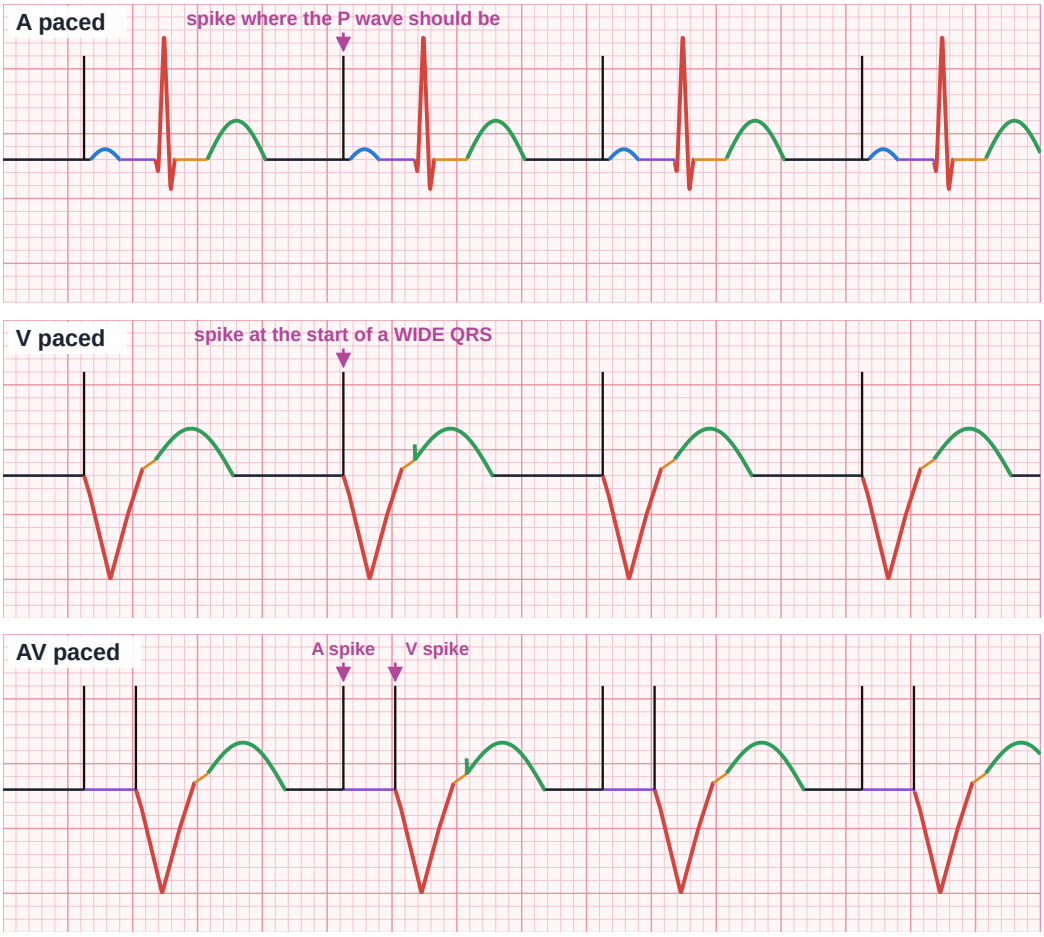
Capture = did the stimulus make the heart depolarize? Staff turn the strength up (like a TENS unit) until every spike is followed by a waveform.



"Smart" (demand) pacemakers sense the patient's own beats and fire only when too much time passes without one.

3 The pacer spike tells you everything

Big idea: Each time the pacemaker fires, it draws a **sharp vertical line** on the strip. **Where** that spike sits tells you which chamber was paced. "It really is that simple."



Drawn examples (P = blue, PRI = purple, QRS = red, T = green, spikes = black lines).

Spike location	What happened	Name
Where the **P wave** should be, followed by a QRS	The atrial lead stimulated the atria. The impulse then went down the normal path (AV node → His → bundle branches), so the QRS is **narrow**	**A paced**
At the very **beginning of the QRS**; no P wave	The ventricular lead stimulated the ventricle directly, so the QRS is **wide**	**V paced**
Both: one where the P should be AND one at the start of the QRS (2 spikes per beat)	Both leads fired, one after the other	**AV paced**

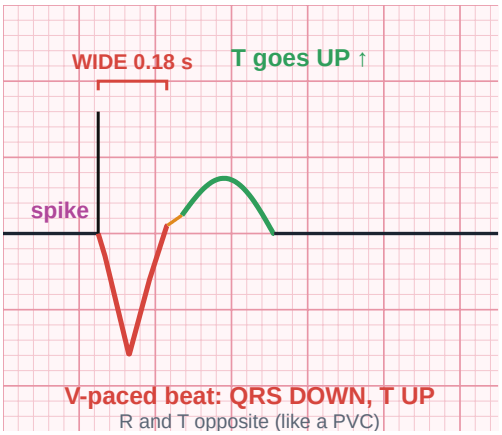
MEMORY TRICK

Count the spikes per beat and note their position. **One spike in front** = A. **One spike on the QRS** = V. **Two spikes** = AV.

4 Why V-paced beats are wide (and look like PVCs)

Big idea: When the wire stimulates the ventricle directly, the impulse doesn't use the fast conduction system. It **spreads cell to cell**, which is slower, so the QRS is **wide**, just like a PVC, and **R and T point in opposite directions**.



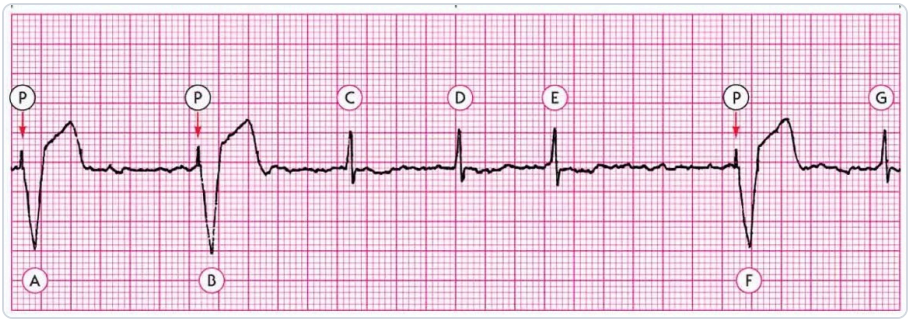


INSTRUCTOR, 0:10:36

"Any time you see a beat with a **spike** with a **wide QRS**, **R and T going in opposite directions**, you know that is a ventricular paced beat."

★ Paced beats mixed with the patient's own beats

Big idea: A pacemaker often **supplements** the patient's own rhythm. It fires only when needed, so some beats are paced and others aren't.



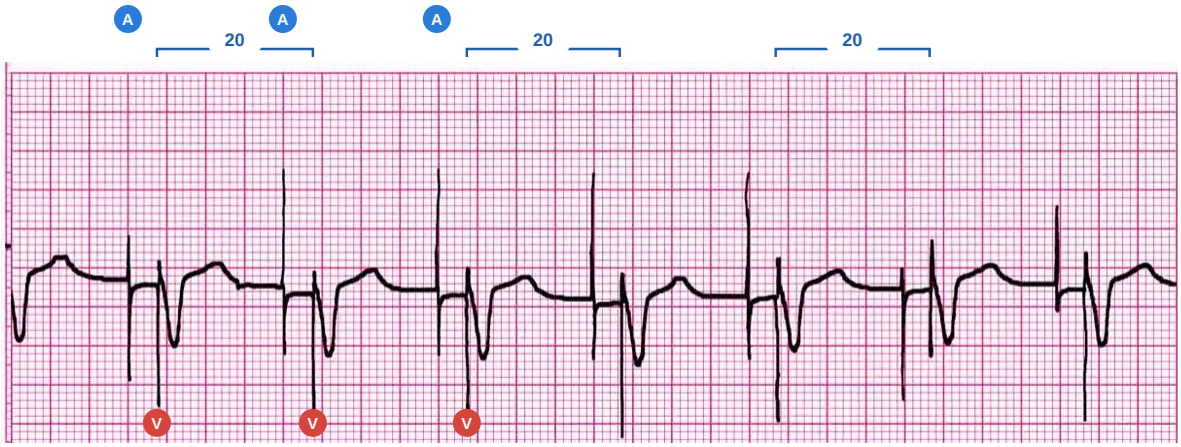
Beats A, B and F have a pacer spike and a wide QRS (ventricular paced). Beats C, D, E and G are the patient's own narrow beats. The underlying rhythm is **atrial fibrillation**: no P waves, f waves, and irregular spacing (18 vs 15 boxes).

GOOD NEWS (0:12:17)

"I'm not going to do that to you." On the homework and test, **the entire rhythm will be paced**, not some beats and not others.

5 AV paced: worked strip

Big idea: There are **two spikes per beat**: one where the P wave should be (atria) and one at the start of the QRS (ventricles). Write **"AV paced"**.



The class strip (0:12:27). A = atrial spike, V = ventricular spike. The QRS points down and the T points up (opposite directions).

Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (20 boxes)	1500 ÷ 20 = **75**	**Paced** (spike where the P should be)	Atrial spike → start of QRS (here the V spike) = 4 boxes = **0.16 s**	About 3 boxes or a bit more (≈0.12 s), **wide**, R and T opposite	**AV paced**

MEASURING THE PRI ON A PACED STRIP

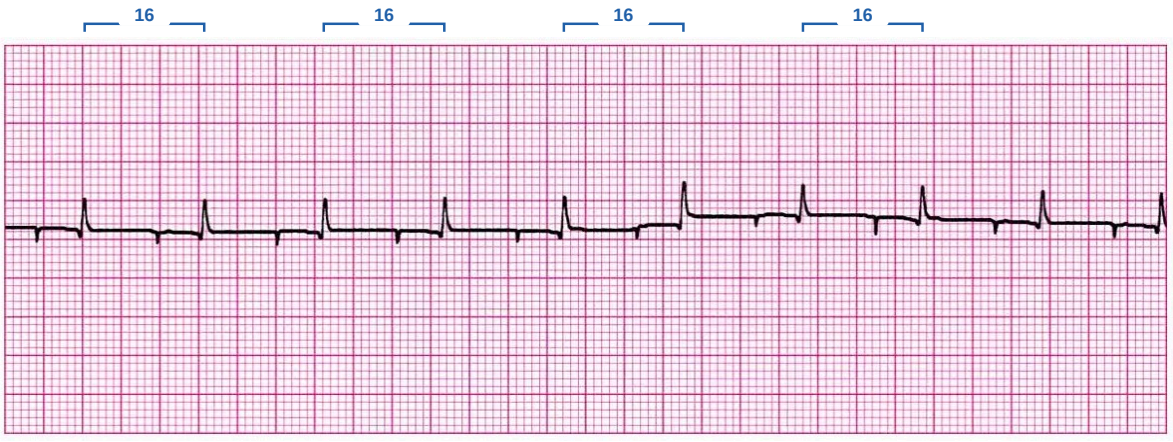
Measure from the **atrial pacer spike** to the **beginning of the QRS**. On an AV paced strip that is "spike to spike". Don't let the spikes that go both up and down confuse you.

SLIDE TITLE VS. INSTRUCTOR

The textbook slide calls this strip "Pacemaker Rhythms (Ventricular)", but the instructor points out **two spikes per beat** and interprets it as **AV paced**. Go with his reading.

6 A paced: worked strip

Big idea: The spike is only where the **P wave** should be. Below the atria conduction is normal, so the QRS is **narrow** and there's **no spike at the QRS**. Write **"A paced"**.



"Atrial Pacemaker with 100% Capture" (0:18:50): a small atrial spike, a long PRI to a small negative Q, then a positive R, with no spike near the QRS. R-to-R ≈ 16 boxes.

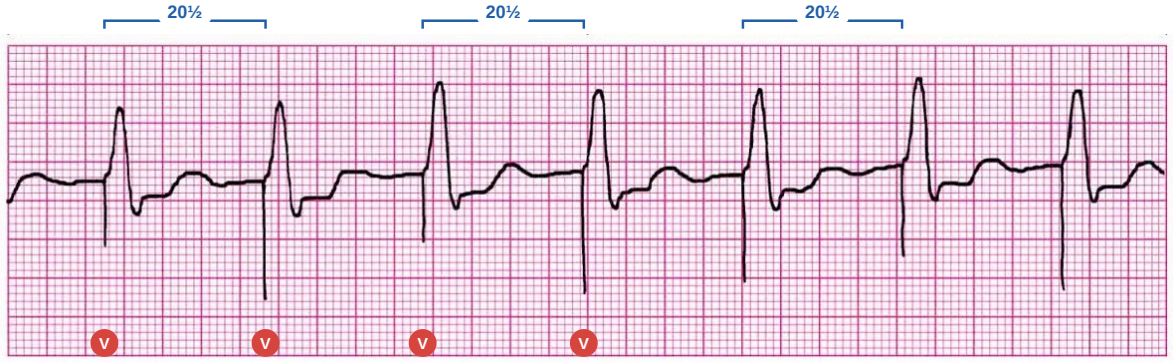
Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (16 boxes)	1500 ÷ 16 = **94**	**Paced**	Spike → start of QRS = 5½ boxes = **0.22 s**	1½ boxes = **0.06 s** (narrow)	**A paced**

STUDENT QUESTION (0:24:40)

A student asked how to tell atrial pacing from ventricular pacing. His answer: look at **where** the spike is. An A-paced spike sits where the P wave belongs, separated from the QRS by a PRI. A V-paced spike sits right at the start of the QRS, and those QRSs are wide.

7 V paced: worked strip

Big idea: The spike is at the **beginning of a wide QRS**. There are **no P waves** and **no atrial spike**, so P waves = "none" and PRI = "none". Write **"V paced"**.



"Ventricular Pacemaker Showing 100% Capture" (0:22:01): T wave → isoelectric line → spike → QRS → T wave, repeating.

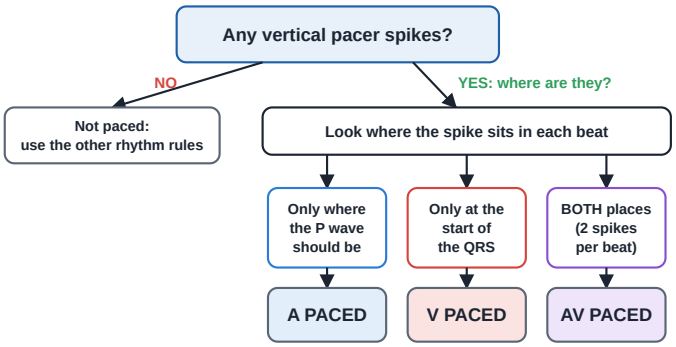
Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (20½ boxes)	1500 ÷ 20.5 = **73**	**None**	**None**	From the spike to the end of the small dip after the QRS = 5 boxes = **0.20 s**	**V paced**

INSTRUCTOR, 0:27:32

"V pacing is the **only time** I'm not going to have P waves and I'm not going to have a PRI." On A paced and AV paced strips, write **"paced"** for P waves and still measure the PRI.

★ Cheat sheet: paced rhythms

	A paced	V paced	AV paced
Spike location	Where the P wave should be	Beginning of the QRS	Both (2 spikes per beat)
Regularity	Regular (all paced rhythms he gives you are regular) → rate = 1500 ÷ small boxes		
P waves	"Paced"	**None**	"Paced"
PRI	Atrial spike → start of QRS	**None**	Atrial spike → start of QRS (spike to spike)
QRS	Narrow (normal conduction)	Wide; R and T opposite	Wide; R and T opposite
Class strip	94 bpm · PRI 0.22 · QRS 0.06	73 bpm · no PRI · QRS 0.20	75 bpm · PRI 0.16 · QRS ≈0.12



Decision tree: find the spike, then find where it sits.

MEMORY HOOKS

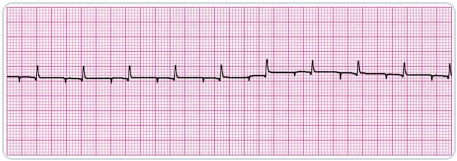
- **Spike = paced.** No spike, not paced.
- **A**trial spike sits in the P wave's "seat". **V**entricular spike is glued to the QRS.
- Wide + spike + R/T opposite = **V**-paced beat (looks like a PVC).
- Only V paced gets "none / none" for P and PRI.

ON THE EXAM

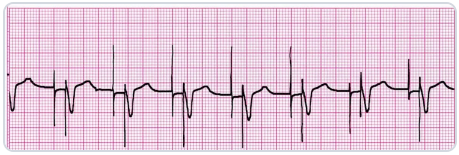
Expect A, V or AV paced only, and fully paced strips. **AV paced is guaranteed** on the test and on the final.

? Practice quiz

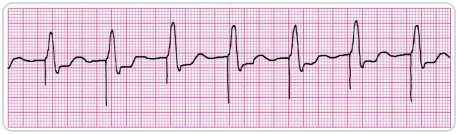
1. What single feature on the strip tells you a rhythm is paced?
2. A spike sits where the P wave should be and is followed by a narrow QRS, with no spike at the QRS. Name the rhythm.
3. Why is the QRS of a ventricular paced beat wide?
4. Which paced rhythm is the only one where you write "none" for both P waves and PRI?
5. On an AV paced strip, where do you start and stop when measuring the PRI?
6. What do you write in the P wave blank for an A paced or AV paced rhythm?
7. R-to-R is 16 small boxes on a regular paced strip. What is the rate?
8. Name strips X, Y and Z below. Give the rate and the reason for your interpretation.
9. On strip Z, how wide is the QRS in seconds (5 small boxes)?
10. On the "paced complexes" strip (section ★), which beats are paced, and what is the underlying rhythm?
11. (Background) What does "capture" mean?
12. Give the two main reasons a patient needs a pacemaker.



Strip X



Strip Y



Strip Z

Answers. 1) A sharp vertical pacer spike. 2) A paced. 3) The impulse starts in the ventricle and spreads cell to cell instead of through the fast conduction system, so it's slow and wide, like a PVC. 4) V paced. 5) From the atrial pacer spike to the beginning of the QRS (spike to spike), e.g. 4 boxes = 0.16 s. 6) "Paced." 7) 1500 ÷ 16 ≈ 94 bpm. 8) X: **A paced**, 94 (spike only where the P belongs, narrow QRS). Y: **AV paced**, 75 (two spikes per beat, wide QRS). Z: **V paced**, 73 (spike at the start of a wide QRS, no P waves). 9) 5 × 0.04 = 0.20 s. 10) A, B and F are ventricular paced. C, D, E and G are the patient's own beats. The underlying rhythm is atrial fibrillation. 11) The stimulus actually made the heart depolarize (every spike is followed by a waveform). 12) The heart's own pacemaker (the SA node) is too slow to sustain life, or there's complete (third-degree) heart block with a slow lower pacemaker.